



MATHEMATICS
HSSC-I
SECTION – A (Marks 20)
Time allowed: 25 Minutes

Candidate Sign. _____

Invigilator Sign. _____

Answer Sheet No. _____

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Version No.				
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Deleting/overwriting is not allowed.
Do not use lead pencil.

Fill the relevant bubble against each question according to curriculum:

Question	A	B	C	D	A	B	C	D
1. Period of $\tan \frac{5x}{7}$ is:	$\frac{5\pi}{7}$	5π	$\frac{7\pi}{5}$	7π	☐	☐	☐	☐
2. A function is periodic if:	$f(x) = f(-x)$	$f(x) = 1$	$f(x) = 0$	$f(x+T) = f(x)$	☐	☐	☐	☐
3. Range of $a + b \sin 2\theta$ is:	$a \pm b$	$a + b$	$a - b$	$2(a \pm b)$	☐	☐	☐	☐
4. Number of ways of arrangements of the letters in the word "SUNDAY" are:	120	480	600	720	☐	☐	☐	☐
5. Number of ways to select 8 players from 12 are:	495	720	11885	19958	☐	☐	☐	☐
6. Factorize $z^2 + 289$:	$(z+17)(z-17)$	$\begin{matrix} (z+17i) \\ (z+17i) \end{matrix}$	$\begin{matrix} (z+17i) \\ (z-17i) \end{matrix}$	$\begin{matrix} (z-17i) \\ (z-17i) \end{matrix}$	☐	☐	☐	☐
7. Argument of $z = \sqrt{3} + i$ is:	$\frac{\pi}{3}$	$\frac{\pi}{4}$	$\frac{\pi}{9}$	$\frac{\pi}{6}$	☐	☐	☐	☐
8. Rank of matrix $A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 7 \\ 0 & 0 & 0 \end{bmatrix}$ is:	1	2	3	4	☐	☐	☐	☐
9. A consistent system MUST have at least:	At least one solution	Infinite solutions	No solution	Maximum two solutions	☐	☐	☐	☐
10. Formula of n^{th} term of an A.P is:	$a + nd$	$a + (n-1)d$	$a + (n+1)d$	$a(n-1)d$	☐	☐	☐	☐
11. H.P decreases when A.P:	Decreases	Increases	Constant	Zero	☐	☐	☐	☐



	Question	A	B	C	D	A	B	C	
12.	Binomial expansion uses:	Factorization	Roots	Division	Pascal triangle	☐	☐	☐	☐
13.	In mathematical induction assume true for $n = k$ is:	Base case	Proof	Hypothesis	Result	☐	☐	☐	☐
14.	In polynomial division Degree of remainder is:	Greater	Equal	Less than divisor	Infinite	☐	☐	☐	☐
15.	If $(x - 2)$ is factor of $f(x)$, then $f(2) =$	-2	0	1	2	☐	☐	☐	☐
16.	Scalar product of two non-zero vector is maximum when both vectors are:	Parallel	In opposite direction	Mutually perpendicular	At 45° to each other	☐	☐	☐	☐
17.	If vector $\vec{u} = 2\hat{i} + 3\hat{j} - 4\hat{k}$ and $\vec{v} = -2\hat{i} - \hat{k}$ then the dot product of $\vec{u}$ and $\vec{v}$ will be:	6	5	0	-2	☐	☐	☐	☐
18.	Three vectors $\vec{P}$, $\vec{Q}$ and $\vec{R}$ are coplanar if:	$\vec{P} \times (\vec{Q} \times \vec{R}) = 0$	$(\vec{P} \cdot \vec{Q}) \times \vec{R} = 0$	$(\vec{P} \cdot \vec{Q}) \cdot \vec{R} = 0$	$\vec{P} \cdot (\vec{Q} \times \vec{R}) = 0$	☐	☐	☐	☐
19.	$\cos(A - B) =$	$\cos A \cos B - \sin A \sin B$	$\cos A \cos B + \sin A \sin B$	$\cos A \sin B + \sin A \cos B$	$\cos A \sin B - \sin A \cos B$	☐	☐	☐	☐
20.	Using product to sum or difference formula $\sin 60^\circ \sin 30^\circ =$	$\frac{\sqrt{3}}{4}$	$\frac{\sqrt{3}}{2}$	1	-1	☐	☐	☐	☐

—1HA-I 26008 (B) (30081)—

ROLL NUMBER

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MATHEMATICS HSSC-I

60 89

Time allowed: 2:35 Hours

Total Marks Sections B and C: 80

SECTION - B (Marks 48)

Q. 2 Solve the following parts.

(12 x 4 = 48)

(Use of the graph paper is not necessary. Candidates can make their own grid on answer book)

(i)	Find the sum of the geometric series $\frac{2}{3}, -1, \frac{3}{2}, \dots$ up to 5 terms.	04	OR	Sketch the graph of $y = 2\cos x$ for $-\pi \leq x \leq \pi$	04
(ii)	Find the last two digits of $(12)^{11}$.	04	OR	Find the rank of matrix: $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}$	04
(iii)	Given $z_1 = 2(\cos 30^\circ + i\sin 30^\circ)$ and $z_2 = 3(\cos 60^\circ + i\sin 60^\circ)$, find $z_1 z_2$ in polar form.	04	OR	Find the value of 'n' if ${}^nP_4 : {}^{n-1}P_3 = 9:1$	04
(iv)	Find the remainder when 9^{205} is divided by 31 using Binomial expansion.	04	OR	Find all roots of: $x^3 - 3x^2 - 4x + 12$ by Factor theorem.	04
(v)	If $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$ then find A_{13}, A_{23}, A_{31} and A_{33} .	04	OR	Prove by mathematical induction that: $2^n \geq n+1$ for all $n \geq 1$	04
(vi)	Solve the following equations: $(2+i)\omega = 1$ $(3-i)z + 5\omega = 2-i$	04	OR	A force $\vec{F} = 3\hat{i} - 2\hat{j} + \hat{k}$ acts at point (1,2,3). Find the moment about the origin.	04
(vii)	Verify the identity: $\cos 3\alpha = 4\cos^3 \alpha - 3\cos \alpha$	04	OR	A class has 5 boys and 4 girls. In how many ways can a group of 3 students be formed?	04
(viii)	Find the volume of the parallelepiped: $\vec{a} = \hat{i} + \hat{j} + 2\hat{k}, \vec{b} = 2\hat{i} + \hat{j} - \hat{k}, \vec{c} = \hat{i} + \hat{k}$	04	OR	Find the maximum and minimum values of: $y = \frac{1}{4 + 5\cos 3x}$	2+2
(ix)	Determine whether $(x+2)$ is a factor of: $f(x) = x^3 + 2x^2 - 4x - 8$	04	OR	Find the sum to n terms of the sequence: 3, 9, 27, 81...	04
(x)	Determine the properties of the graph of: $y = \cos x$ i.e. Domain, Range, Period and Amplitude.	04	OR	Find the value of " λ " so that vectors $\vec{v} = \hat{i} - \hat{j} + \hat{k}, \vec{u} = 2\hat{i} + \hat{j} + \hat{k}$ and $\vec{w} = -\hat{i} + \lambda\hat{j} + 2\hat{k}$ are coplanar.	04
(xi)	Find the Sum of first 20 terms of A.P with $a=1, d=3$.	04	OR	Verify: $\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$	04
(xii)	Find the inverse of matrix if possible: $\begin{bmatrix} 0 & 3 & 4 \\ -4 & 2 & 6 \\ 2 & 0 & 5 \end{bmatrix}$	04	OR	Convert $\cos \frac{2\theta + 2\phi}{2} \sin \frac{2\theta - 2\phi}{2}$ into sum or difference form:	04

SECTION - C (Marks 32)

Note: Solve the following Questions.

(4 x 8 = 32)

(Use of the graph paper is not necessary. Candidates can make their own grid on answer book)

Q.3	a. Recognize the sequence (as A.P or G.P) and find the general term of the sequence $4, 2, 1, \frac{1}{2}, \dots$ b. Find the sum up to n terms. c. Find the sum up to infinity.	4+2 +2	OR	A committee of 5 members is to be selected from a group of 8 men and 6 women. In how many ways can the committee be formed if it must include at least 2 women?	08
Q.4	Solve: $2x + y - z = 1$ $x + 2y + z = 4$ $3x + y + 2z = 7$	08	OR	In a robotics project, a mechanical arm moves in such a way that its vertical position h (in meters) is modeled by $h(t) = 15\cos\left(\frac{t}{2}\right)$, where t is time in seconds. Determine the first time t when the vertical position of the arm reaches 7.5 meters. Also determine the value of t for maximum height.	08
Q.5	Find the middle term(s) in the expansion of $\left(3x^2 - \frac{1}{2x}\right)^{10}$. Also find the 9 th term.	08	OR	If $\sin \alpha = \frac{3}{5}, \sin \beta = \frac{1}{\sqrt{5}}$, where terminal ray of α and β are in 2 nd quadrant, then find the value of $\cos(\alpha - \beta)$ and $\sin(\alpha + \beta)$.	08
Q.6	a. For what value of k the remainder of $2x^3 - kx^2 + 16x + 12$ will be zero if the polynomial, is divided by $x - 2$ b. Factorize the polynomial completely.	3+5	OR	a. Prove that scalar triple product is invariant under cyclic permutation. b. Verify using vectors $\vec{i}, \vec{j}, \vec{k}$	5+3



MATHEMATICS
HSSC-I
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Candidate Sign. _____

Invigilator Sign. _____

Answer Sheet No. _____

Signature should not cross this line

Version No.				
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ROLL NUMBER						

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Deleting/overwriting is not allowed.
Do not use lead pencil.

Fill the relevant bubble against each question according to curriculum:

Question	A	B	C	D	A	B	C	D
Factorize $z^2 + 169$:	$(z+13)(z-13)$	$(z+13i)$ $(z-13i)$	$(z-13i)$ $(z-13i)$	$(z+13i)$ $(z+13i)$	☐	☐	☐	☐
Argument of $z = 1 + i$ is:	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	☐	☐	☐	☐
Rank of identity matrix is:	Less than its order	Equal to its order	Greater than its order	Undefined	☐	☐	☐	☐
A system is inconsistent if it has:	No solution	Unique solution	Infinite solutions	Repeated solutions	☐	☐	☐	☐
5 th term of A.P for which $(a_1 = 2, d = 3)$ is:	14	11	13	15	☐	☐	☐	☐
If an A.P. = $a_1, a_2, a_3, \dots$ then H.P. =	$a_1, a_2, a_3, \dots$	$a_1 + a_2 + a_3 + \dots$	$\frac{1}{a_1 + a_2 + a_3 + \dots}$	$\frac{1}{a_1}, \frac{1}{a_2}, \frac{1}{a_3}, \dots$	☐	☐	☐	☐
Coefficient of middle term:	is always 1	is independent of n	Depends on n	is always zero	☐	☐	☐	☐
In mathematical induction first step is:	Hypothesis	Result	Conclusion	Base case	☐	☐	☐	☐
If a cubic polynomial is divided by a linear polynomial (containing same variable), the degree of remainder will be:	0	1	2	3	☐	☐	☐	☐
Using factor theorem If $f(a) = 0$ then:	$f(a)$ is a factor	a is factor	$(x-a)$ is factor	$(x+a)$ is factor	☐	☐	☐	☐
If dot product of two non-zero vectors is zero, then both vectors are:	parallel	equal	mutually perpendicular	in opposite direction	☐	☐	☐	☐
If $\vec{a} = a_1\hat{i} + b_1\hat{j}$ and $\vec{b} = a_2\hat{i} + b_2\hat{j}$ then $\vec{a} \cdot \vec{b}$ is equal to:	$a_1a_2 + b_1b_2$	$a_1a_2 - b_1b_2$	$a_1b_1 + a_2b_2$	$a_2b_1 - a_1b_2$	☐	☐	☐	☐



	Question	A	B	C	D	A	B	C	D
13.	Two vectors $\vec{a}$ and $\vec{b}$ will be orthogonal, if the angle θ between them is:	π	$\frac{\pi}{2}$	2π	$\frac{2\pi}{3}$	☐	☐	☐	☐
14.	$\cos(\pi + \beta)$ is equal to:	$\cos \beta$	$-\cos \beta$	$\sin \beta$	$-\sin \beta$	☐	☐	☐	☐
15.	$\cos A \cos B$ is equal to:	$\frac{1}{2}[\cos(A+B) + \cos(A-B)]$	$\cos(A+B)$	$\cos(A-B)$	$\frac{1}{2}[\sin(A+B) + \sin(A-B)]$	☐	☐	☐	☐
16.	Period of $\cos \frac{2x}{5}$ is:	$\frac{5}{2}\pi$	2π	$\frac{2}{5}\pi$	5π	☐	☐	☐	☐
17.	Which of the given functions is even?	$f(x) = x^3$	$f(x) = \sin x$	$f(x) = x$	$f(x) = \cos x$	☐	☐	☐	☐
18.	Maximum value of $y = 3 + 2\sin \theta$ is:	5	4	3	2	☐	☐	☐	☐
19.	Number of ways of arrangement of the letters in the word "GARDEN" are:	480	600	720	840	☐	☐	☐	☐
20.	Number of ways to choose 10 members from 15 are:	360,360	$\frac{15!}{10!}$	$\frac{15!}{5!}$	3003	☐	☐	☐	☐

—1HA-I 26008-32081 (D)—

ROLL NUMBER						





MATHEMATICS HSSC-I

62

Time allowed: 2:35 Hours

Total Marks Sections B and C: 80

SECTION – B (Marks 48)

Q. 2 Solve the following parts.

(12 x 4 = 48)

(Use of the graph paper is not necessary. Candidates can make their own grid on answer book)

(i)	Solve the system: $z + 2\omega = 4i$ $2z + 5\omega = 3$	04	OR	A force $\vec{F} = 2i + 3j + k$ moves an object through displacement $\vec{d} = i + 2j + 2k$. find the work done.	04
(ii)	Verify that $\sin 4\theta = 8\sin\theta\cos^3\theta - 4\sin\theta\cos\theta$	04	OR	Evaluate $\frac{(n-1)!}{(n-3)!}$	04
(iii)	Find the volume of a tetrahedron formed by vectors: $\vec{a} = 2i + j, \vec{b} = i + 3j + k, \vec{c} = j + k$	04	OR	Find the maximum and minimum values of: $y = \frac{1}{2 + 3\sin(2x + 1)}$	04
(iv)	Find the value(s) of K if $x - 1$ is a factor of: $f(x) = x^3 + K^2 - 2K + 1$	04	OR	Find the sum of 6 terms of sequence: $1 + \frac{1}{2} + \frac{1}{4} + \dots$	04
(v)	Without drawing graph determine the amplitude, period and range of: $y = 2\sin(3x + 1)$	04	OR	Find the angle between vectors: $\vec{a} = 4i + j - 8k, \vec{b} = i + 4j + k$	04
(vi)	Find the 10 th term of an A.P with $a = 3$ and the 7 th term of it is 15.	04	OR	Using fundamental law of trigonometry find the value of $\cos 15^\circ$. (do not use calculator)	04
(vii)	Find the rank of matrix: $\begin{bmatrix} 3 & -4 & 2 \\ 2 & 3 & 5 \\ 1 & 0 & 1 \end{bmatrix}$	04	OR	Express $2\sin 12\theta \cos 6\theta$ as a sum or difference.	04
(viii)	The 3 rd term of a geometric sequence is 16 and the 5 th term is 64. Find the common ratio and 1 st term. Also write the sequence for positive values.	04	OR	Find the inverse of given matrix if possible: $\begin{bmatrix} 2 & 1 & 0 \\ 0 & 3 & 2 \\ 2 & 1 & 4 \end{bmatrix}$	04
(ix)	Determine the last two digits of $(13)^{10}$.	04	OR	Sketch the graph of $y = 3\sin x$ for $-\pi \leq x \leq \pi$	04
(x)	Convert $z = 1 + i$ into polar form and then find z^2 using polar representation.	04	OR	Find r if ${}^{54}P_r : {}^{56}P_{r+3} = 1:30800$	04
(xi)	Use Binomial Theorem to find the remainder when 8^{205} is divided by 48.	04	OR	Factorize the polynomial: $x^3 - 4x^2 + x + 6$	04
(xii)	For what value of λ the given matrix is singular? $\begin{bmatrix} 1 & -1 & 2 \\ 2 & 1 & \lambda \\ 3 & 1 & 2 \end{bmatrix}$	04	OR	Use mathematical induction to prove: $1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$	04

SECTION – C (Marks 32)

Note: Solve the following Questions.

(4 x 8 = 32)

(Use of the graph paper is not necessary. Candidates can make their own grid on answer book)

Q.3	a. A company's profit over the last few years can be modeled by the polynomial $P(x) = x^3 + 6x^2 - x - 6$. If x represents years since 2010, calculate the company's profit in 2012 by dividing $P(x)$ and $x - 2$ finding the remainder. b. Verify your answer by long division.	08	OR	If $\vec{a} = 3\hat{i} - 2\hat{j} + 4\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$, then a. Find $\vec{a} \times \vec{b}$ b. Find $(\vec{a} \times \vec{b}) \cdot \vec{a}$ and verify that $\vec{a} \times \vec{b} \perp \vec{a}$ c. Find $(\vec{a} \times \vec{b}) \cdot \vec{b}$ and verify that $\vec{a} \times \vec{b} \perp \vec{b}$	08
Q.4	Find the term independent of x in the expansion of $\left(2x - \frac{1}{x}\right)^{12}$. Also find the 10 th term.	4+4	OR	Prove fundamental law of trigonometry: $\cos(A - B) = \cos A \cos B + \sin A \sin B$ Diagram is necessary.	08
Q.5	Solve the system: $x + y + z = 6$ $2x + 3y + z = 10$ $x + 2y + 3z = 11$	08	OR	The height of seat on a Ferris wheel is modeled by the equation $h(t) = 10 + 8\sin \frac{\pi}{6}t$ a. Find the period of motion b. Determine the maximum and minimum height c. At what time is the height maximum?	3+2+3
Q.6	a. For the sequence 2, 5, 8, 11, ... find the general term. Also find the sum of first 20 terms. b. Find the sum of the n terms of the series $1.2 + 2.3 + 3.4 + \dots$	4+4	OR	A box contains 7 red, 5 blue and 4 green balls. In how many ways can 5 balls be selected if at least 2 are red?	08